

Topic 6 – Custom Database Lookup Agent

Assessment Implementation & Validation Documentation

Implementation status: Workflow configuration reviewed. The main AI Agent uses Google Gemini instead of OpenAI Chat Model because of the OpenAI API rate-limit issue. The assessment requirements call for a connected Chat Model and do not, in the supplied assessment text, mandate OpenAI specifically.

1. Workflow Architecture

User chat message → Customer Support Agent → customer_lookup Workflow Tool → Customer Lookup Tool → Code-based mock customer database → result returned to Agent → concise customer response.

2. Main AI Agent Configuration

Component	Configuration
Chat Trigger	When Chat Message Received
AI Agent	Customer Support Agent
Chat Model	Google Gemini Chat Model
Model	models/gemini-3.1-flash-lite
Temperature	0.0
Max Output Tokens	512
Max Iterations	5
Memory	Window Buffer Memory; context window 10
Custom Tool	customer_lookup
Required Tool Input	customer_id (string)

3. System Prompt Behavior

- Specific customer questions must invoke customer_lookup before answering.
- Customer IDs follow the CUST001-style format.
- If no customer_id is provided, the Agent asks the user for it.
- If found:false is returned, the Agent clearly states that no customer was found and does not invent details.
- If found:true is returned, the Agent uses only name, email, plan, status, and country.

4. Custom Workflow Tool

The reusable Customer Lookup Tool is triggered by When Called by Agent and receives customer_id. The Lookup Customer Code node normalizes the ID by trimming whitespace and converting it to uppercase, then returns either a matching record or a structured not-found response.

5. Mock Customer Dataset

ID	Name	Plan	Status	Country
CUST001	Alice Johnson	Premium	active	USA

ID	Name	Plan	Status	Country
CUST002	Bob Smith	Basic	active	UK
CUST003	Carla Mendes	Enterprise	suspended	Brazil
CUST004	Daniel Kim	Premium	active	South Korea
CUST005	Ewa Kowalski	Basic	churned	Poland

6. Direct Validation Tests for Loom Demo

Test	Chat Input	Expected Result
1. Successful lookup	Find customer CUST001	Tool is invoked; Alice Johnson record is returned; Agent answers from returned file
2. Another valid ID	What plan does CUST003 have?	Tool is invoked with CUST003; Agent reports Enterprise and can mention suspension
3. Invalid ID	Find customer CUST999	Tool returns found:false; Agent clearly says no customer was found for CUST999;
4. Missing ID	What is the customer's plan?	Agent asks the user to provide a customer_id and does not call the tool without an
5. Case/space handling	Find customer cust002	Tool normalizes input and should return CUST002 / Bob Smith.

7. Loom Demo Flow

- Show the Customer Support AI Agent workflow and its five nodes.
- Open the Google Gemini Chat Model and show model, temperature 0.0, and max output tokens 512.
- Show Window Buffer Memory and max iterations 5 on the Agent.
- Open customer_lookup and show its description plus required customer_id string input.
- Open the Customer Lookup Tool workflow and show the trigger plus Lookup Customer Code node.
- Run the successful CUST001 lookup and show the tool invocation/execution result.
- Run the invalid CUST999 lookup and show found:false handling.
- Run the missing-ID test and show that the Agent requests customer_id.
- Open execution details/logs and briefly show parameter passing from the Agent to the custom tool.

8. Submission Checklist

- Successful customer lookup verified in live n8n
- Invalid customer handling verified in live n8n
- Missing customer_id handling verified
- Execution logs/screenshots captured
- Workflow JSON exported
- Sample dataset included
- README completed
- Public GitHub repository verified
- Loom/YouTube demo recorded

Note: This document records the configured implementation and planned validation. The checklist items above should be marked complete only after the live n8n tests are demonstrated and evidence is captured.